부록3

*파일 2229의 경우 2개의 임무가 하나의 파일에 기록되어 따로 분리하여 분석

*SD 카드 저장 오류로 반드시 시간순과 일치하지 않을 수 있음

파일명 (시간)	Ctrl Mode	Length	Weight	RMSE (deg)	최대각도 (deg)	Sway Energy (deg²·s)	CtrlEffort Energy (cmd²·s)	총임무시간 (0.015~1.985m)(s)
2221	OFF	0.7m	X	6.12	17.82	3531.0	361.97e+06	94.48
2223				6.16	14.61	3582.5	361.85e+06	94.42
2225				5.84	15.99	3216.6	361.85 e+06	94.43
2326	ON	0.7m	X	4.59	14.15	1953.1	369.58 e+06	92.85
2328				6.35	26.87	3538.7	390.07 e+06	87.91
2330				7.09	32.49	4365.2	396.36 e+06	86.80
2226	OFF	1.0m	X	4.30	16.04	2453.1	512.13e+06	132.42
2229_M1				5.08	16.62	3470.4	520.00 e+06	134.41
2229_M2				5.39	17.99	3900.0	520.09 e+06	134.44
2332	ON	1.0m	X	5.40	27.16	3741.5	550.21 e+06	128.17
2335				4.59	20.57	2923.8	530.58 e+06	138.79
2341				3.54	10.94	1699.0	522.96 e+06	135.64
0013	OFF	0.7m	O	5.87	20.80	3247.8	360.87 e+06	94.20
0015				5.79	19.08	3168.9	361.79 e+06	94.44
0017				3.61	10.49	1250.0	362.47 e+06	95.92
2353	ON	0.7m	O	2.43	7.22	562.8	363.77 e+06	94.94
2356				3.92	11.86	1446.4	366.67 e+06	94.33
0028				4.54	12.26	1899.9	373.63 e+06	92.27
0019	OFF	1.0m	O	3.69	11.17	1836.1	521.59 e+06	134.83
0021				3.63	10.14	1771.8	521.41 e+06	134.77
0024				3.52	10.20	1465.4	457.20 e+06	118.54
0000	ON	1.0m	O	1.92	6.65	501.3	523.29 e+06	135.26
0003				2.66	11.92	956.2	520.03 e+06	135.58
0006				2.75	10.94	1016.8	526.20 e+06	134.94

Table 19. 개별 데이터 결과

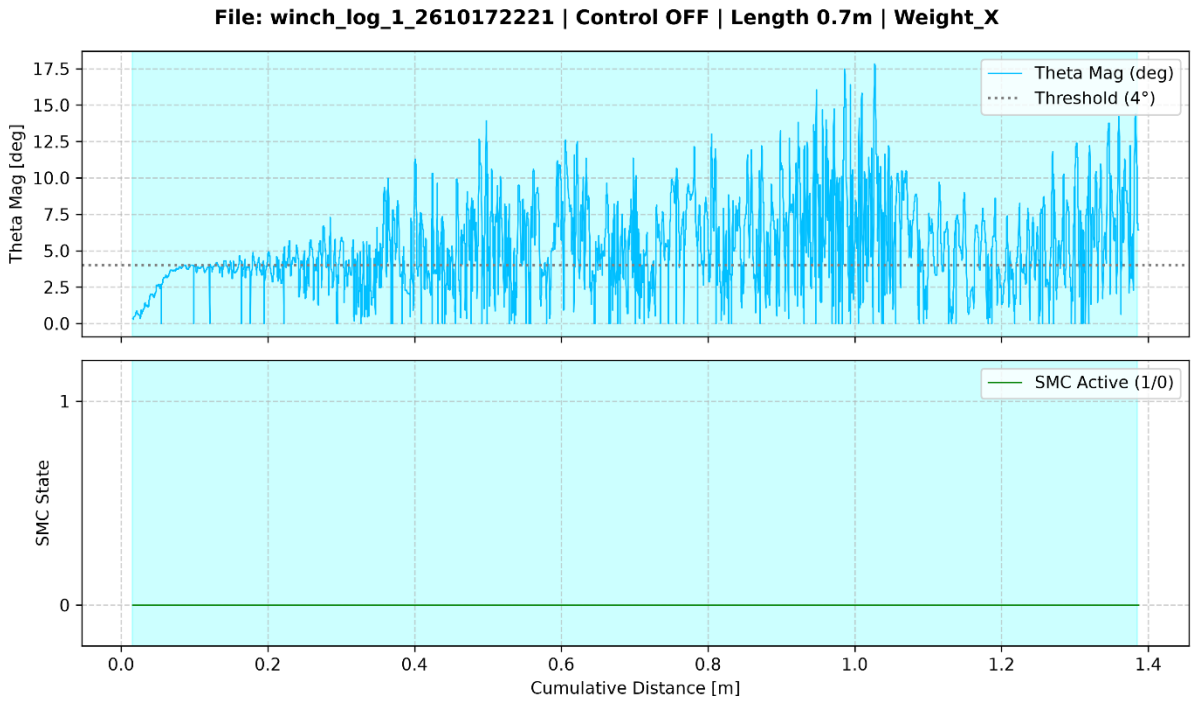

File: winch_log_1_2610172221 | Control OFF | Length 0.7m | Weight_X
Theta Mag [deg]
Theta Mag (deg)
Threshold (4°)
SMC State
SMC Active (1/0)
Cumulative Distance [m]

Fig14

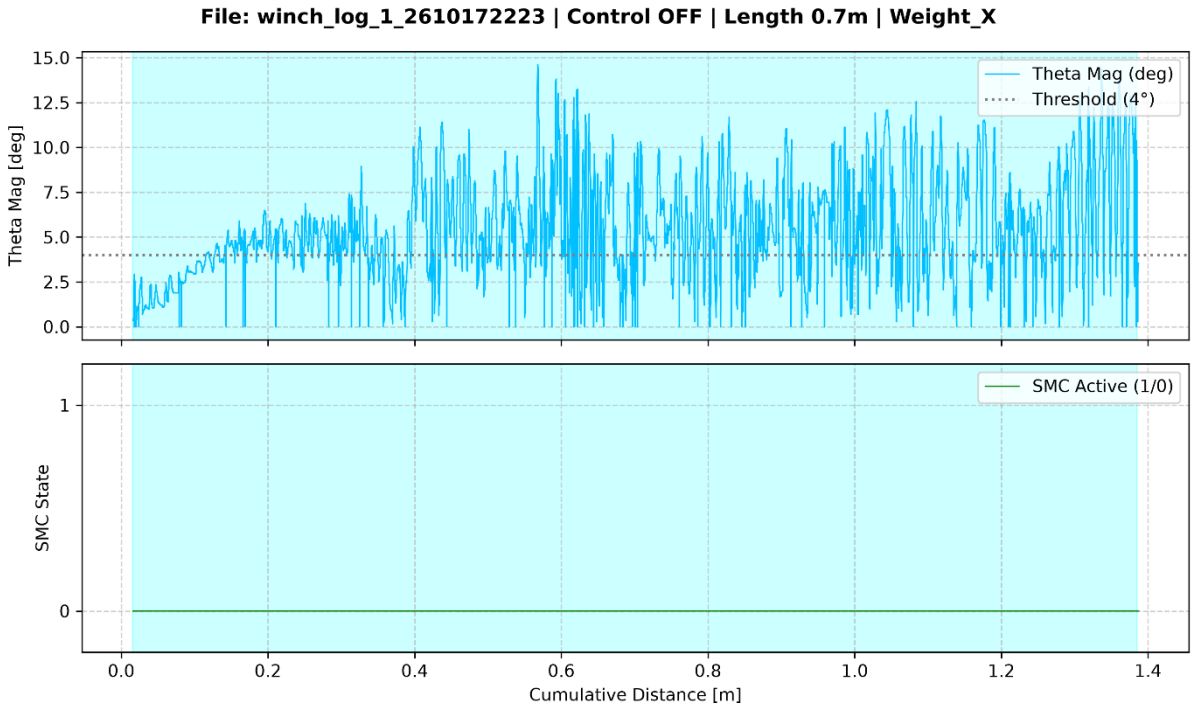

File: winch_log_1_2610172223 | Control OFF | Length 0.7m | Weight_X
Theta Mag [deg]
Theta Mag (deg)
Threshold (4°)
SMC State
SMC Active (1/0)
Cumulative Distance [m]

Fig 15

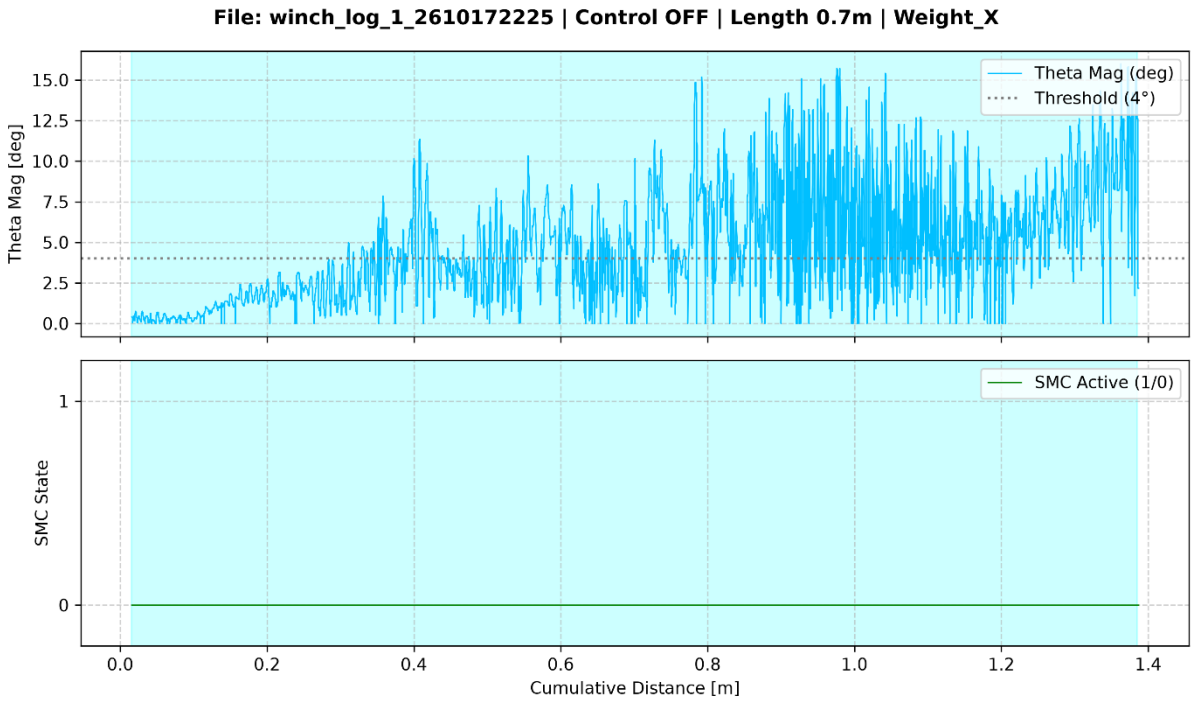

File: winch_log_1_2610172225 | Control OFF | Length 0.7m | Weight_X
Theta Mag (deg)
Threshold (4°)
Theta Mag [deg]
15.0
12.5
10.0
7.5
5.0
2.5
0.0
SMC Active (1/0)
SMC State
1
0
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Cumulative Distance [m]

Fig 16

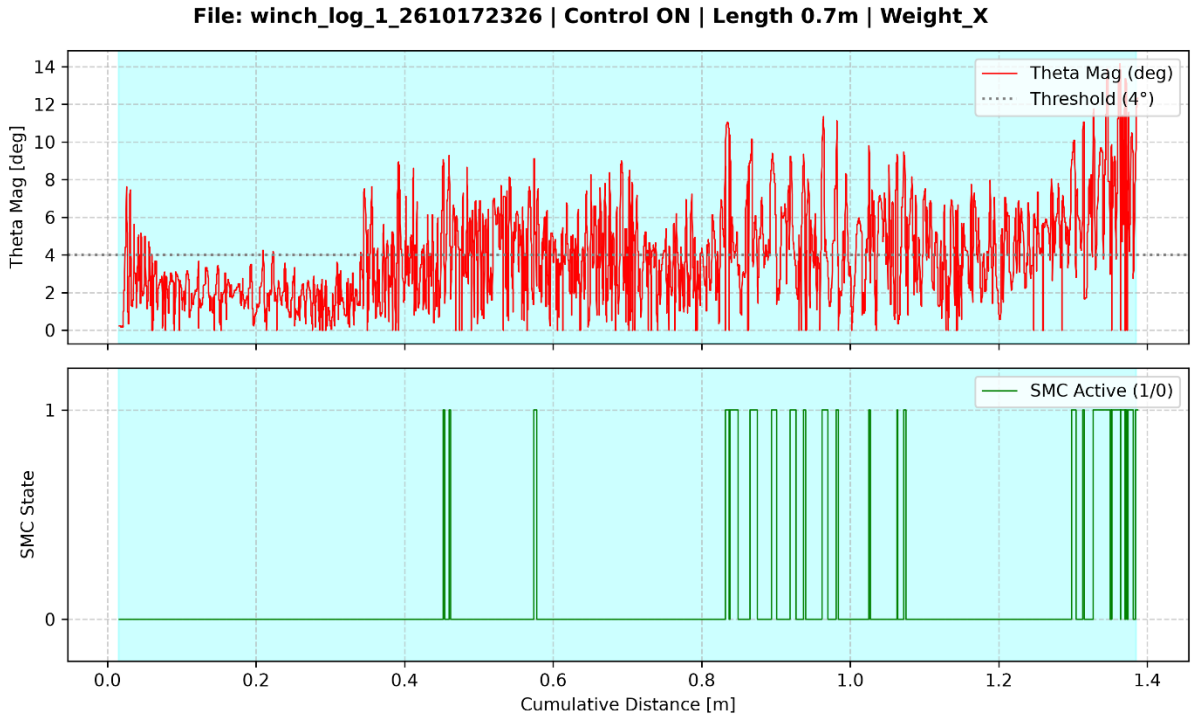

File: winch_log_1_2610172326 | Control ON | Length 0.7m | Weight_X
Theta Mag (deg)
Threshold (4°)
Theta Mag [deg]
14
12
10
8
6
4
2
0
SMC Active (1/0)
SMC State
1
0
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Cumulative Distance [m]

Fig 17

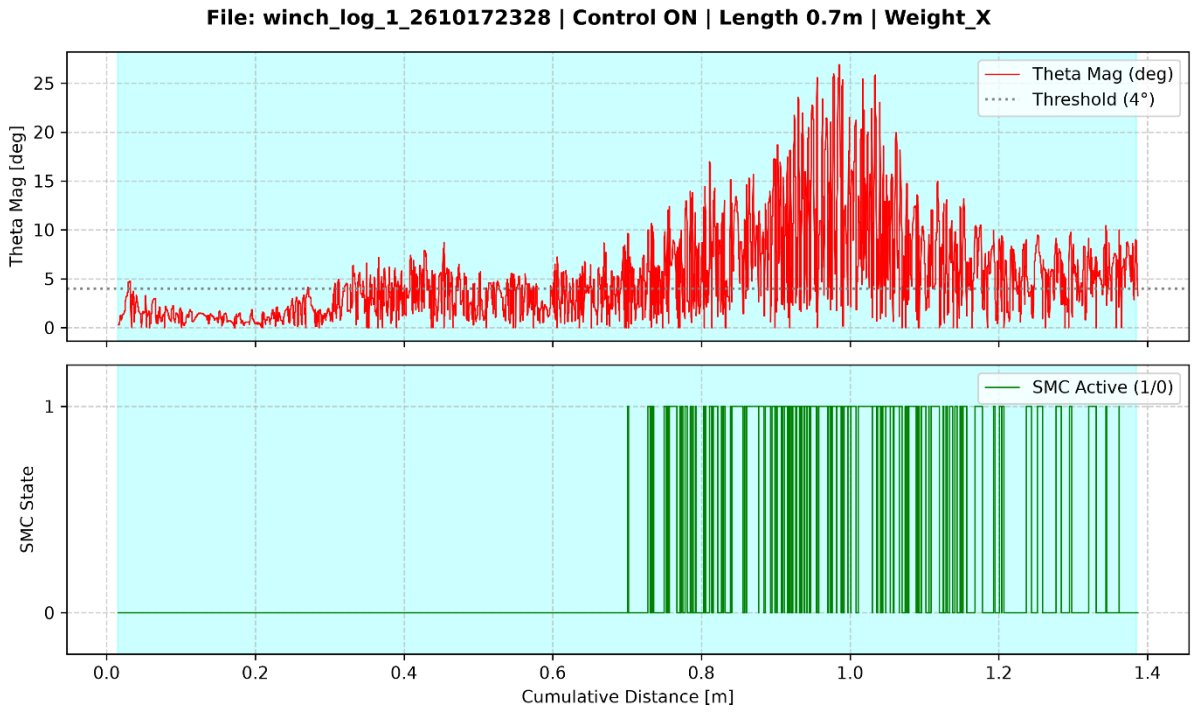

File: winch_log_1_2610172328 | Control ON | Length 0.7m | Weight_X
Theta Mag (deg)
Threshold (4°)
Theta Mag [deg]
SMC Active (1/0)
SMC State
Cumulative Distance [m]

Fig 18

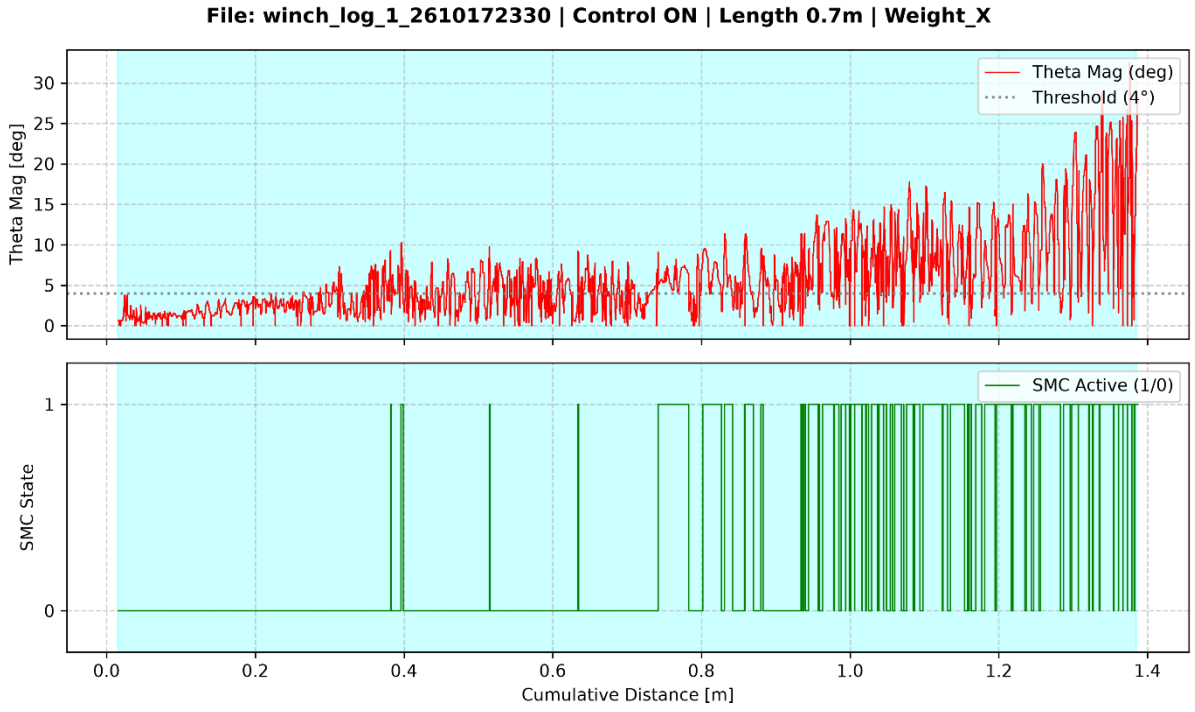

File: winch_log_1_2610172330 | Control ON | Length 0.7m | Weight_X
Theta Mag (deg)
Threshold (4°)
Theta Mag [deg]
SMC Active (1/0)
SMC State
Cumulative Distance [m]

Fig 19

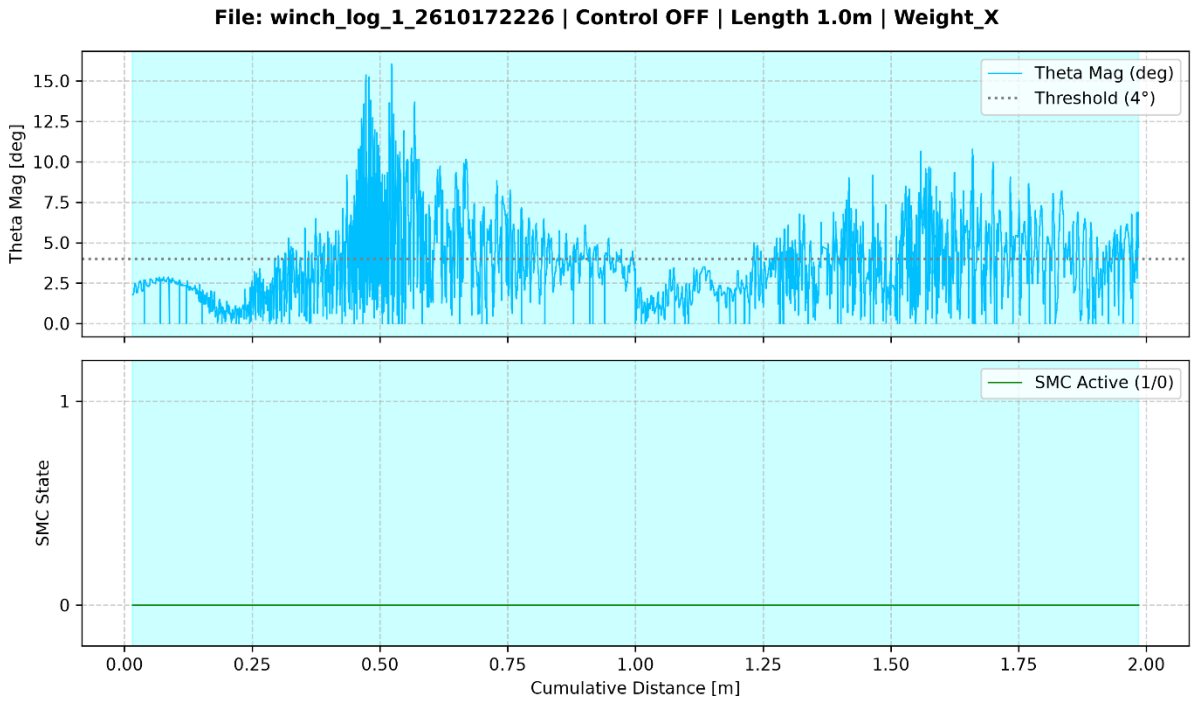

File: winch_log_1_2610172226 | Control OFF | Length 1.0m | Weight_X
Theta Mag (deg)
Threshold (4°)
Theta Mag [deg]
15.0
12.5
10.0
7.5
5.0
2.5
0.0
SMC Active (1/0)
SMC State
1
0
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Cumulative Distance [m]

Fig 20

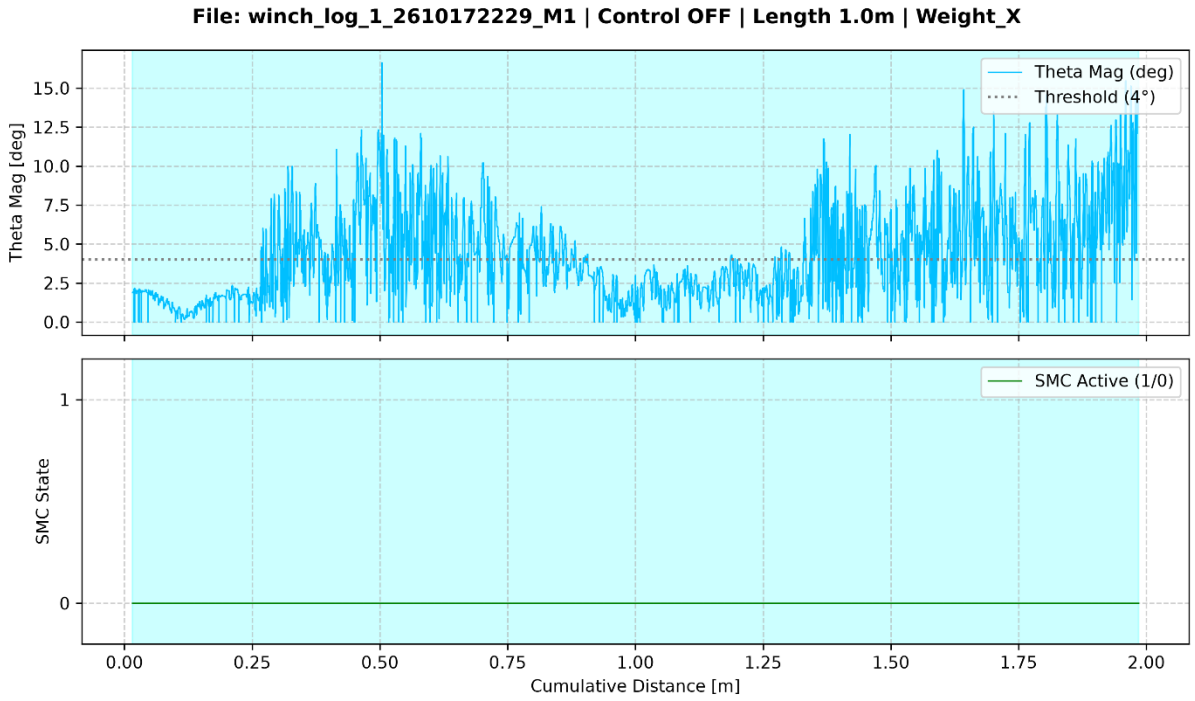

File: winch_log_1_2610172229_M1 | Control OFF | Length 1.0m | Weight_X
Theta Mag (deg)
Threshold (4°)
Theta Mag [deg]
15.0
12.5
10.0
7.5
5.0
2.5
0.0
SMC Active (1/0)
SMC State
1
0
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Cumulative Distance [m]

Fig 21

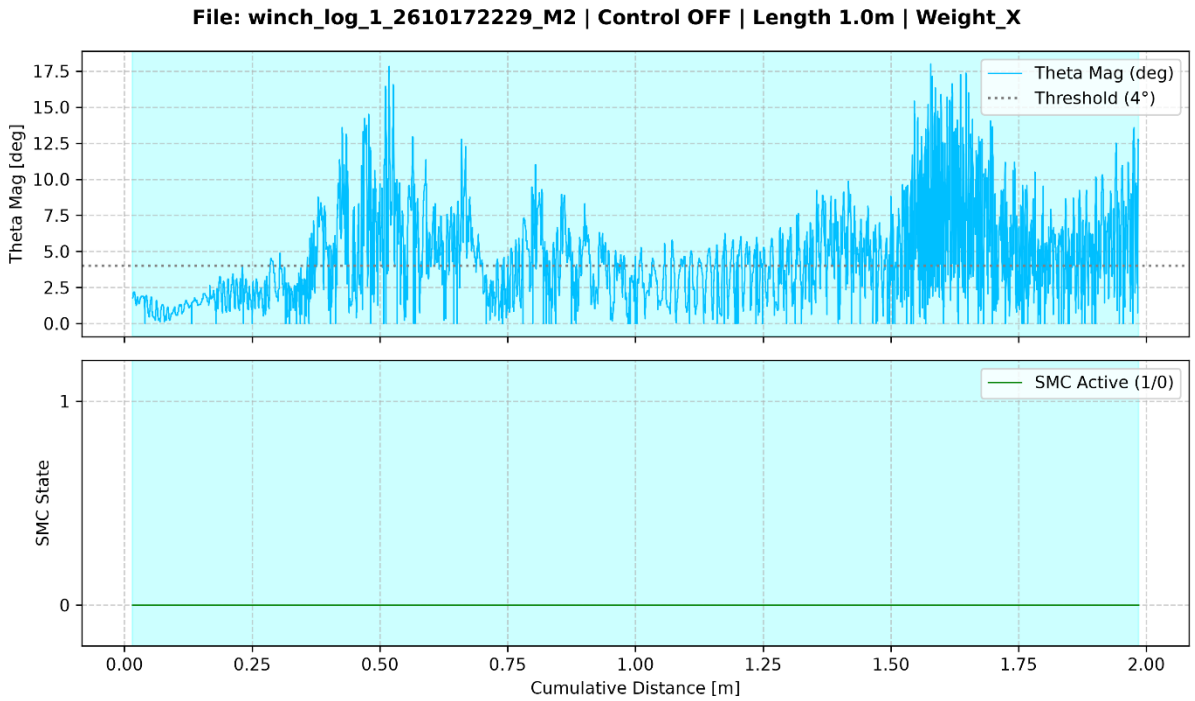

File: winch_log_1_2610172229_M2 | Control OFF | Length 1.0m | Weight_X
Theta Mag (deg)
Threshold (4°)
Theta Mag [deg]
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
SMC Active (1/0)
SMC State
1
0
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Cumulative Distance [m]

Fig22

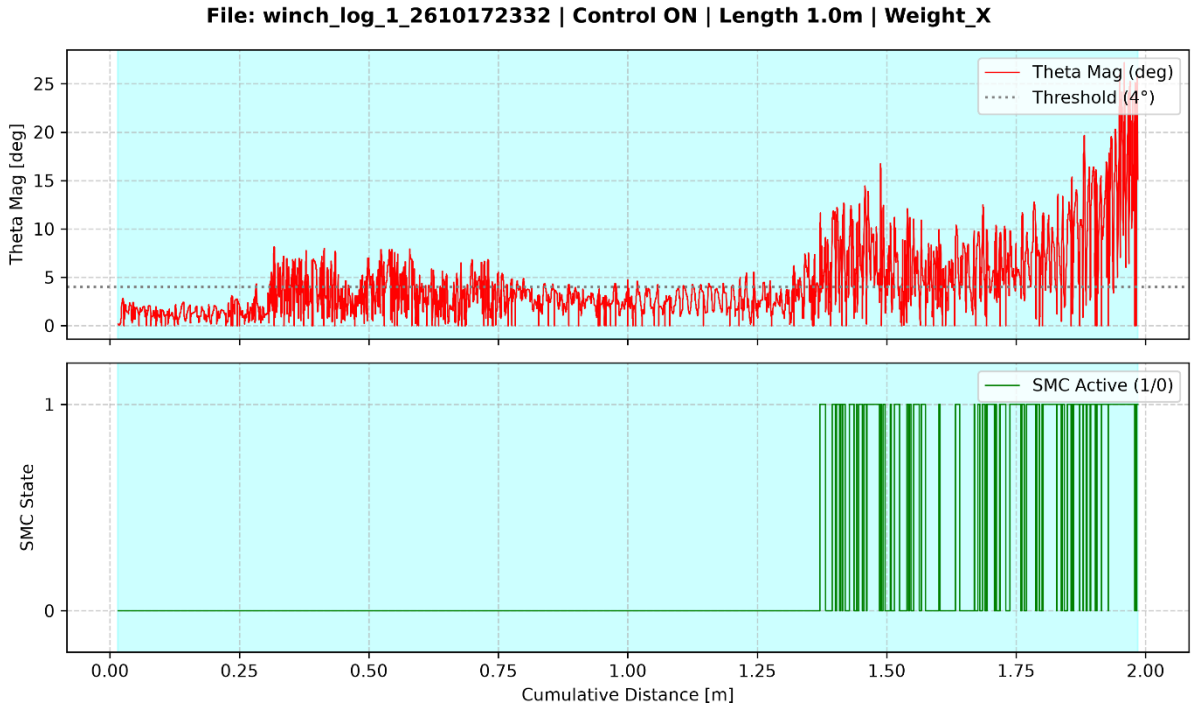

File: winch_log_1_2610172332 | Control ON | Length 1.0m | Weight_X
Theta Mag (deg)
Threshold (4°)
Theta Mag [deg]
25
20
15
10
5
0
SMC Active (1/0)
SMC State
1
0
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Cumulative Distance [m]

Fig 23

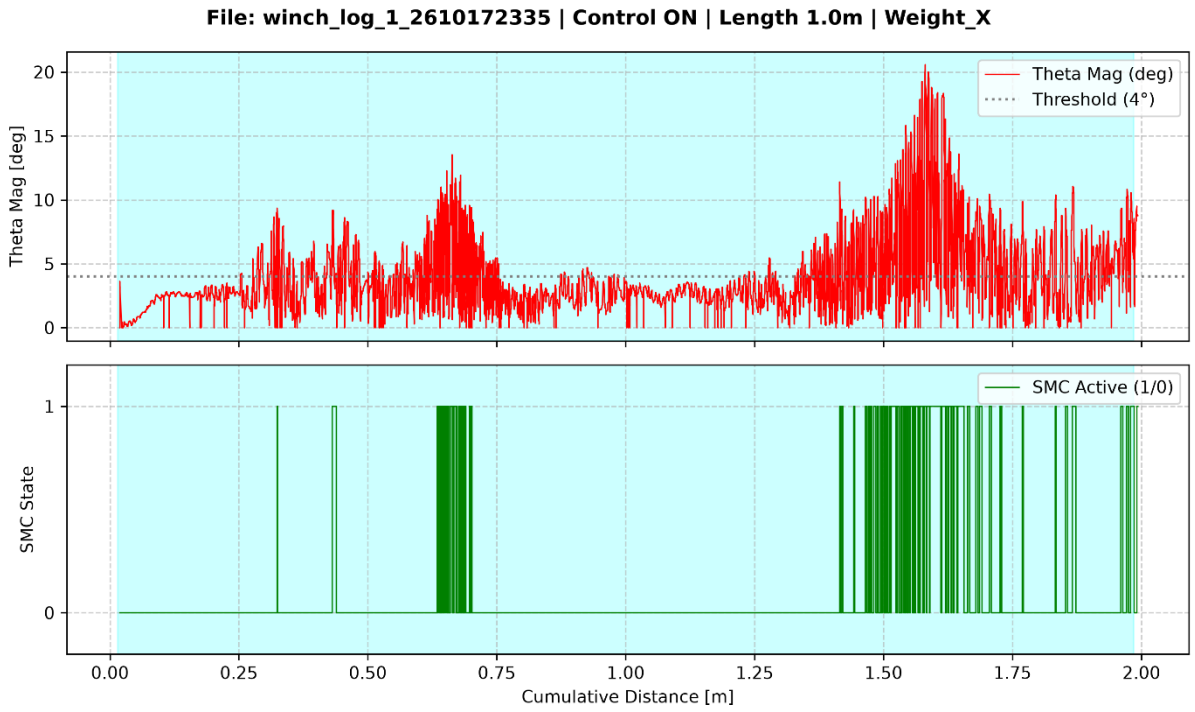

File: winch_log_1_2610172335 | Control ON | Length 1.0m | Weight_X
Theta Mag [deg]
Theta Mag (deg)
Threshold (4°)
SMC State
SMC Active (1/0)
Cumulative Distance [m]

Fig 24

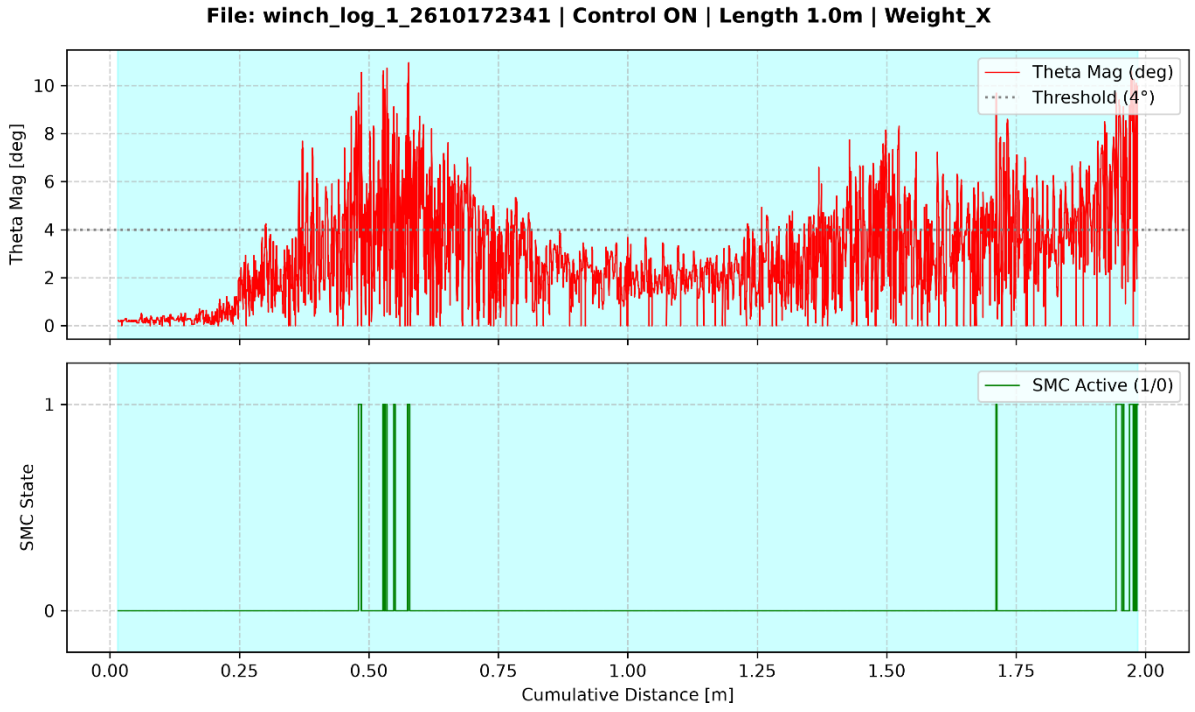

File: winch_log_1_2610172341 | Control ON | Length 1.0m | Weight_X
Theta Mag [deg]
Theta Mag (deg)
Threshold (4°)
SMC State
SMC Active (1/0)
Cumulative Distance [m]

Fig 25

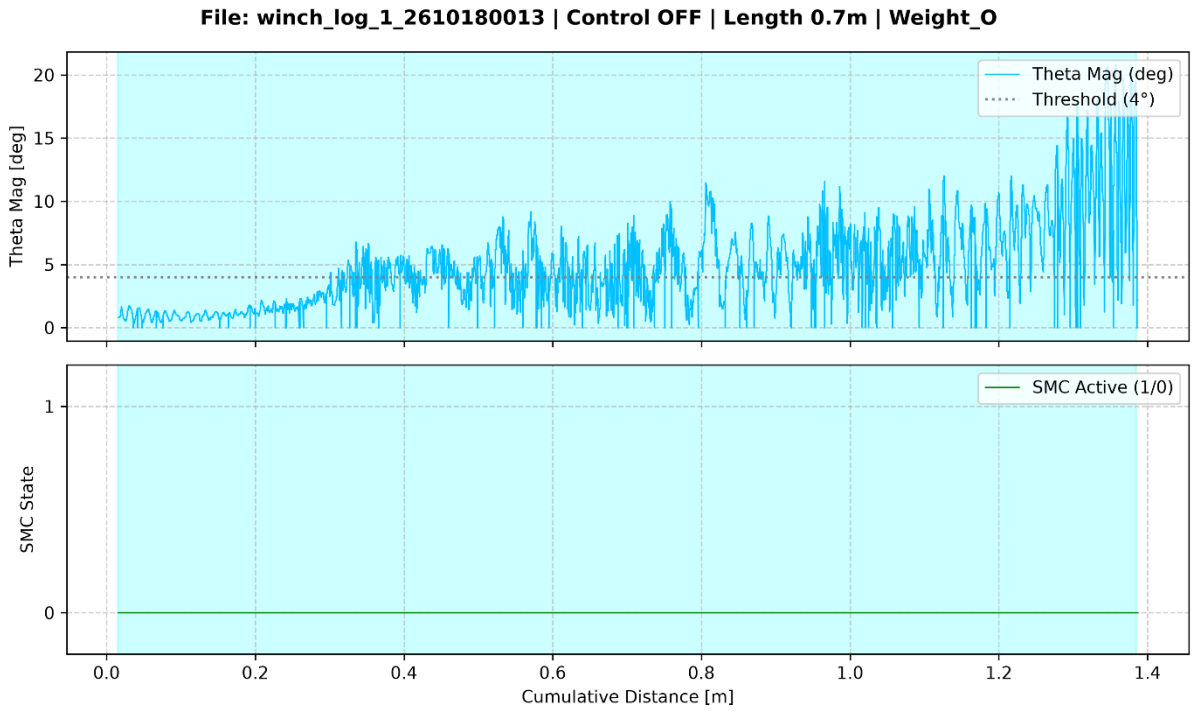

File: winch_log_1_2610180013 | Control OFF | Length 0.7m | Weight_O
Theta Mag [deg]
Theta Mag (deg)
Threshold (4°)
SMC State
SMC Active (1/0)
Cumulative Distance [m]

Fig 26

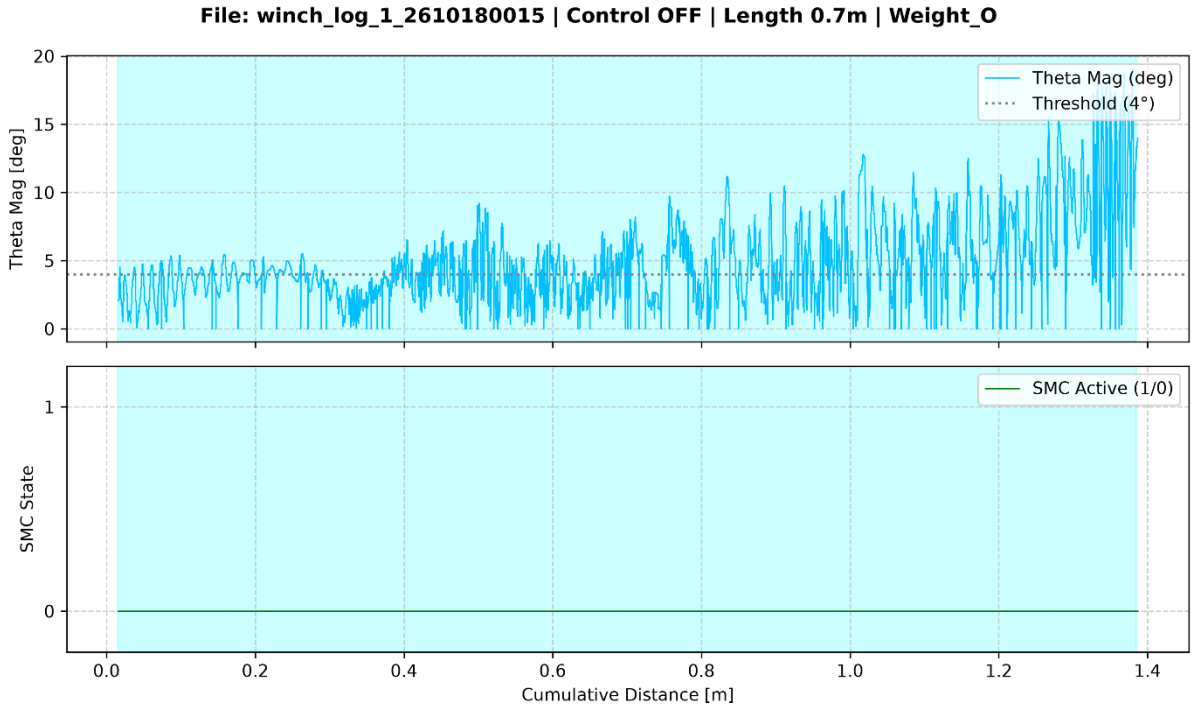

File: winch_log_1_2610180015 | Control OFF | Length 0.7m | Weight_O
Theta Mag [deg]
Theta Mag (deg)
Threshold (4°)
SMC State
SMC Active (1/0)
Cumulative Distance [m]

Fig 27

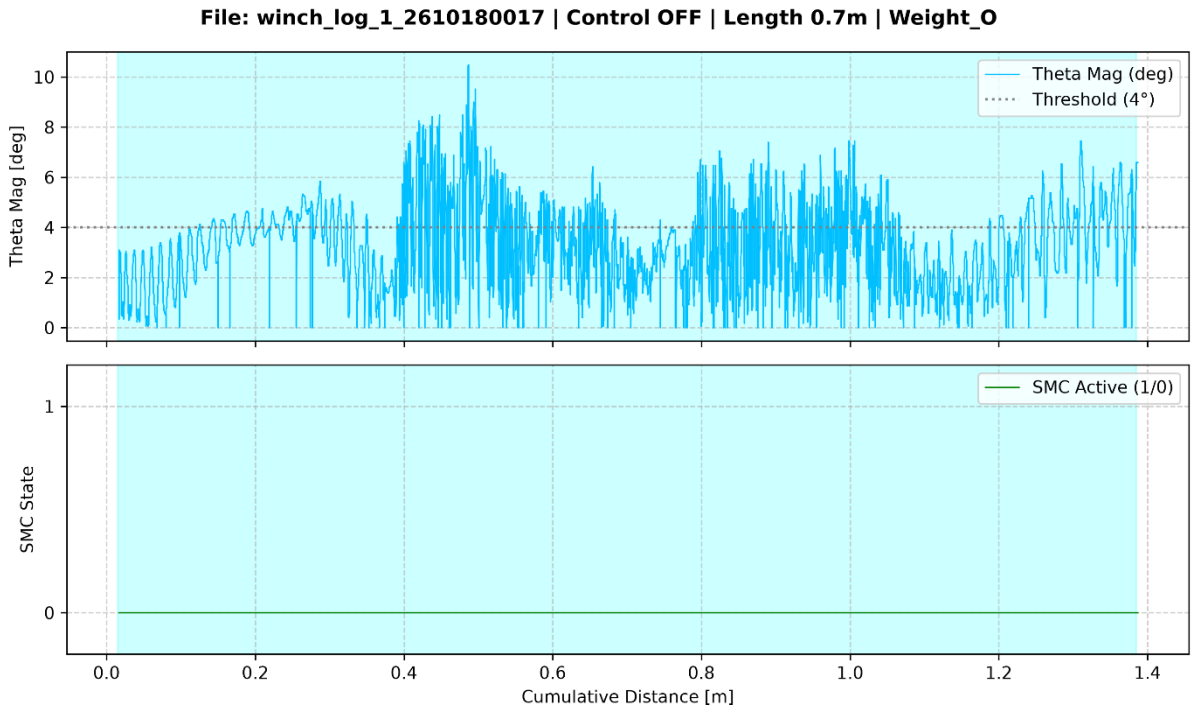

File: winch_log_1_2610180017 | Control OFF | Length 0.7m | Weight_O
Theta Mag [deg]
Theta Mag (deg)
Threshold (4°)
SMC State
SMC Active (1/0)
Cumulative Distance [m]

Fig28

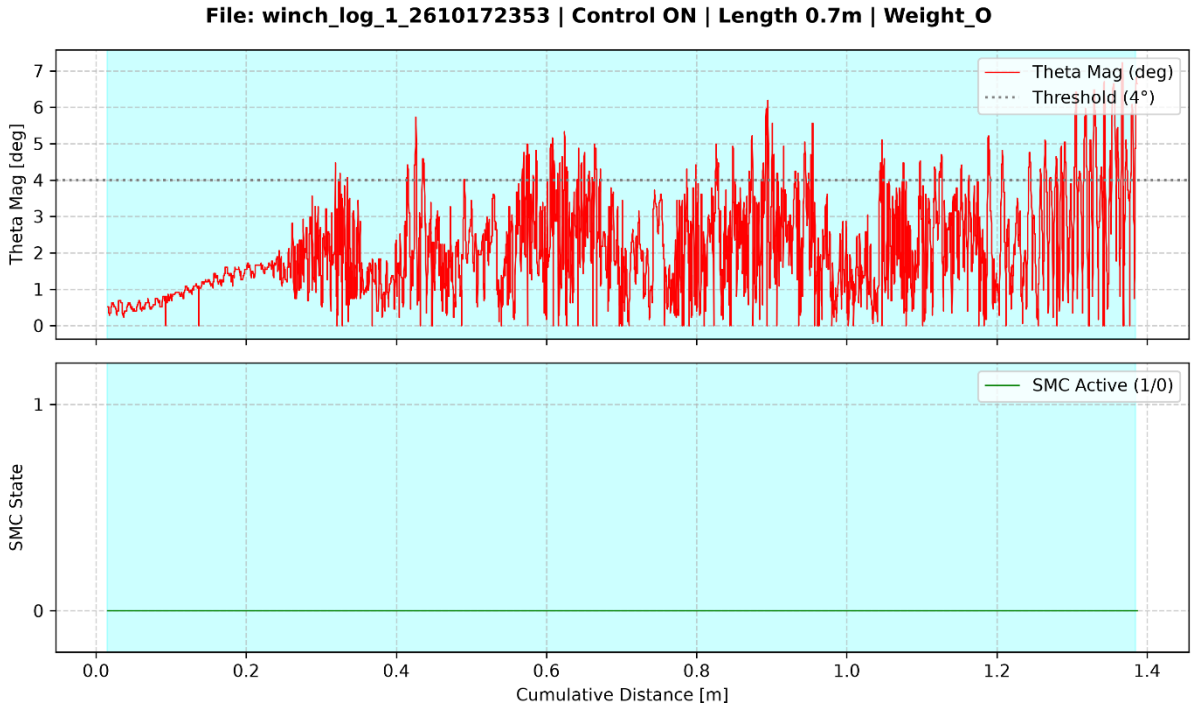

File: winch_log_1_2610172353 | Control ON | Length 0.7m | Weight_O
Theta Mag [deg]
Theta Mag (deg)
Threshold (4°)
SMC State
SMC Active (1/0)
Cumulative Distance [m]

Fig 29

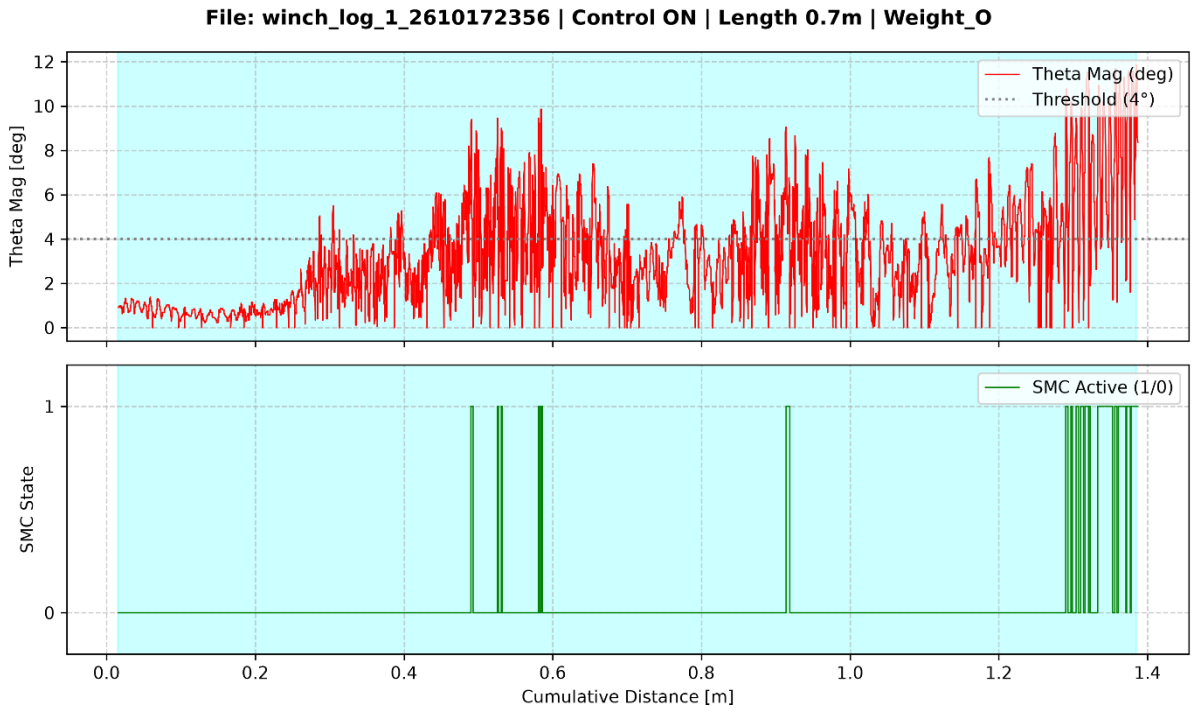

File: winch_log_1_2610172356 | Control ON | Length 0.7m | Weight_O
Theta Mag (deg)
Threshold (4°)
Theta Mag [deg]
SMC Active (1/0)
SMC State
Cumulative Distance [m]

Fig 30

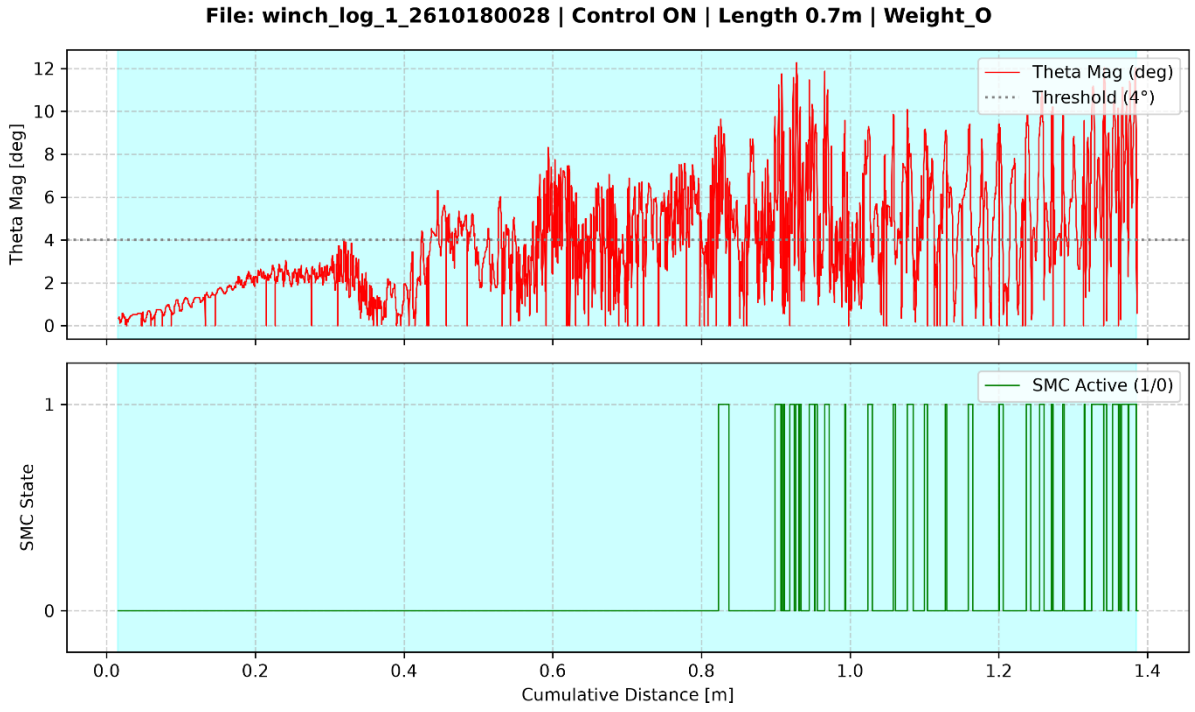

File: winch_log_1_2610180028 | Control ON | Length 0.7m | Weight_O
Theta Mag (deg)
Threshold (4°)
Theta Mag [deg]
SMC Active (1/0)
SMC State
Cumulative Distance [m]

Fig31

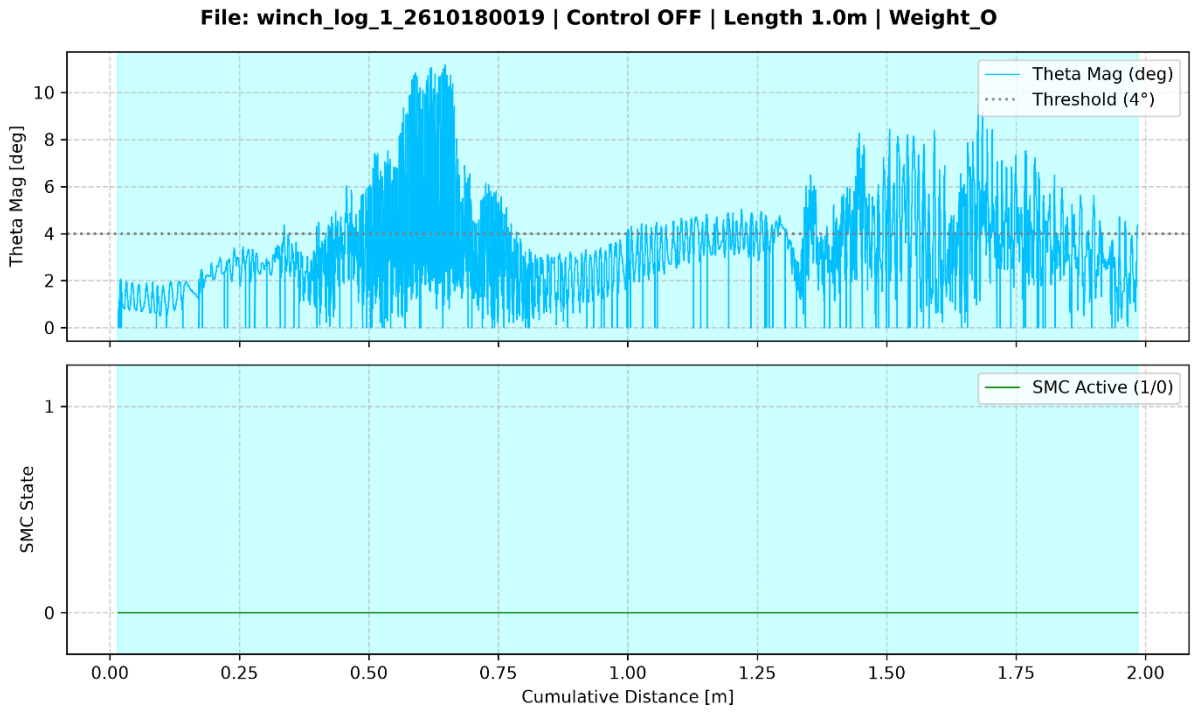

File: winch_log_1_2610180019 | Control OFF | Length 1.0m | Weight_O
Theta Mag (deg)
Threshold (4°)
Theta Mag [deg]
SMC Active (1/0)
SMC State
Cumulative Distance [m]

Fig 32

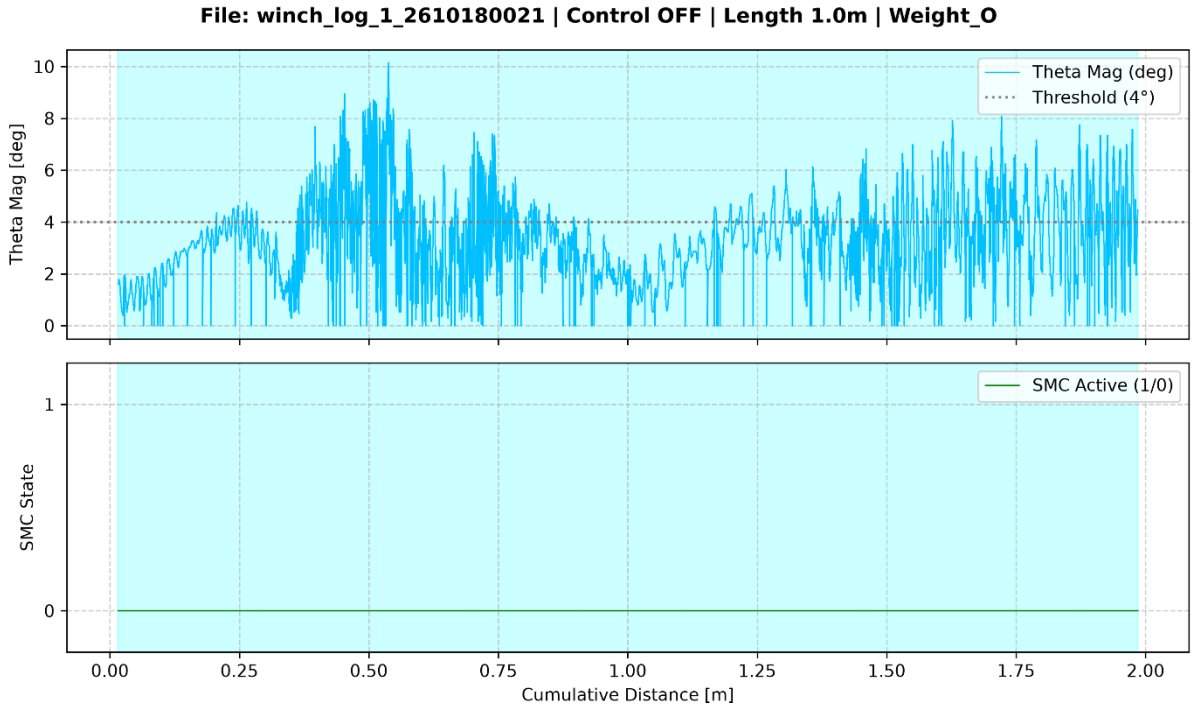

File: winch_log_1_2610180021 | Control OFF | Length 1.0m | Weight_O
Theta Mag (deg)
Threshold (4°)
Theta Mag [deg]
SMC Active (1/0)
SMC State
Cumulative Distance [m]

Fig 33

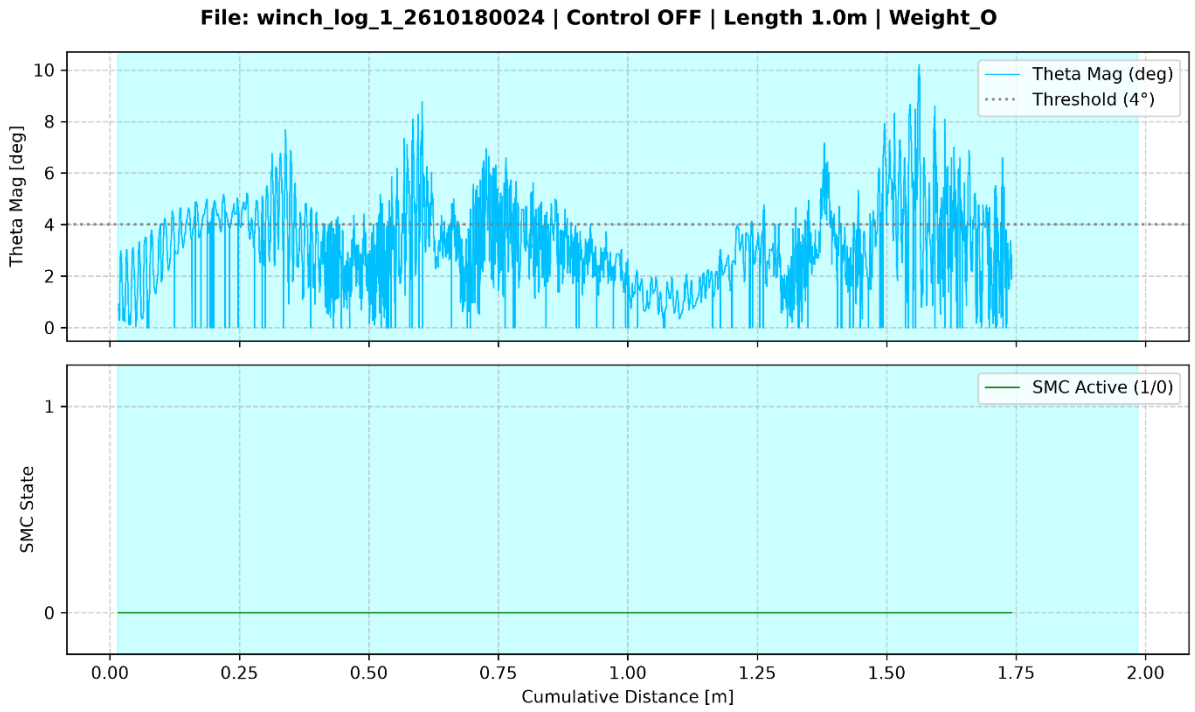

File: winch_log_1_2610180024 | Control OFF | Length 1.0m | Weight_O
Theta Mag [deg]
Threshold (4°)
SMC State
SMC Active (1/0)
Cumulative Distance [m]

Fig34

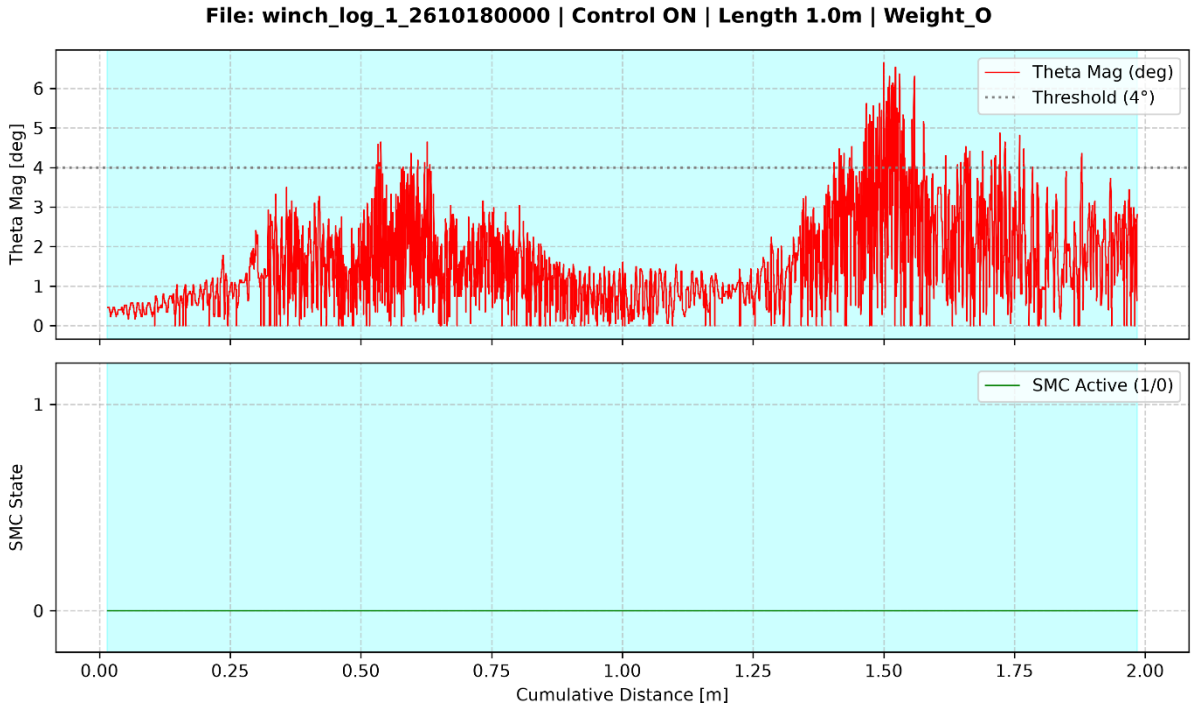

File: winch_log_1_2610180000 | Control ON | Length 1.0m | Weight_O
Theta Mag [deg]
Threshold (4°)
SMC State
SMC Active (1/0)
Cumulative Distance [m]

Fig 35

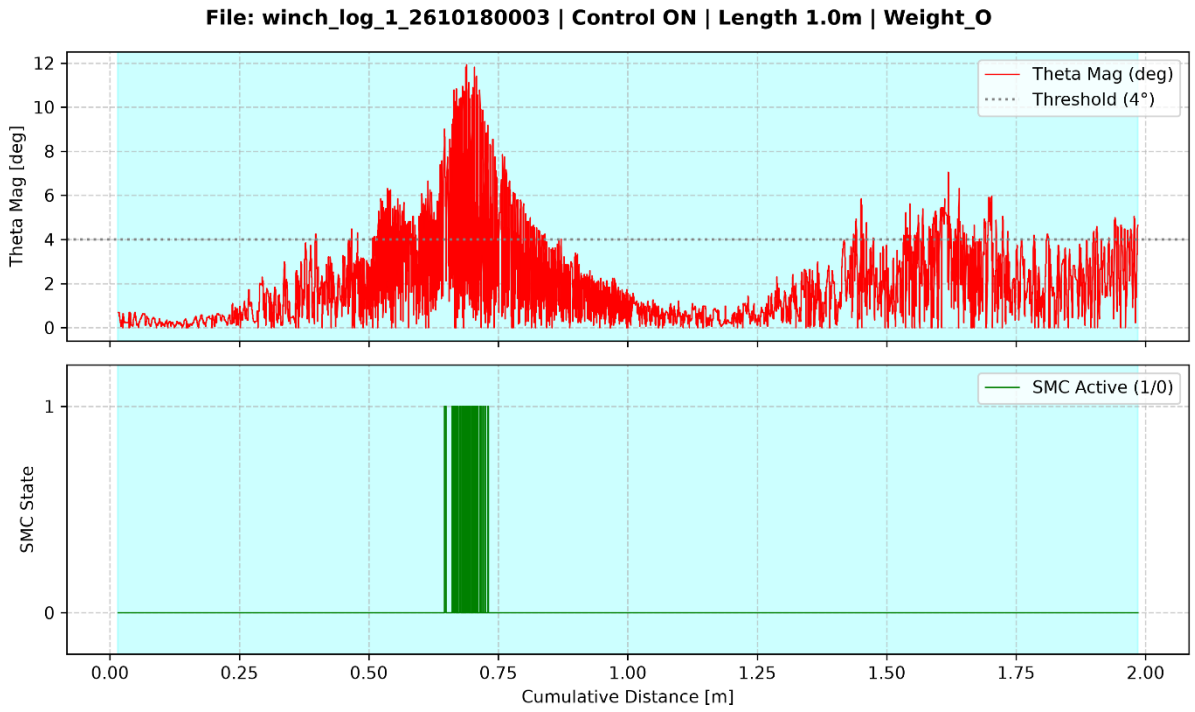

File: winch_log_1_2610180003 | Control ON | Length 1.0m | Weight_O
Theta Mag [deg]
Theta Mag (deg)
Threshold (4°)
SMC State
SMC Active (1/0)
Cumulative Distance [m]

Fig 36

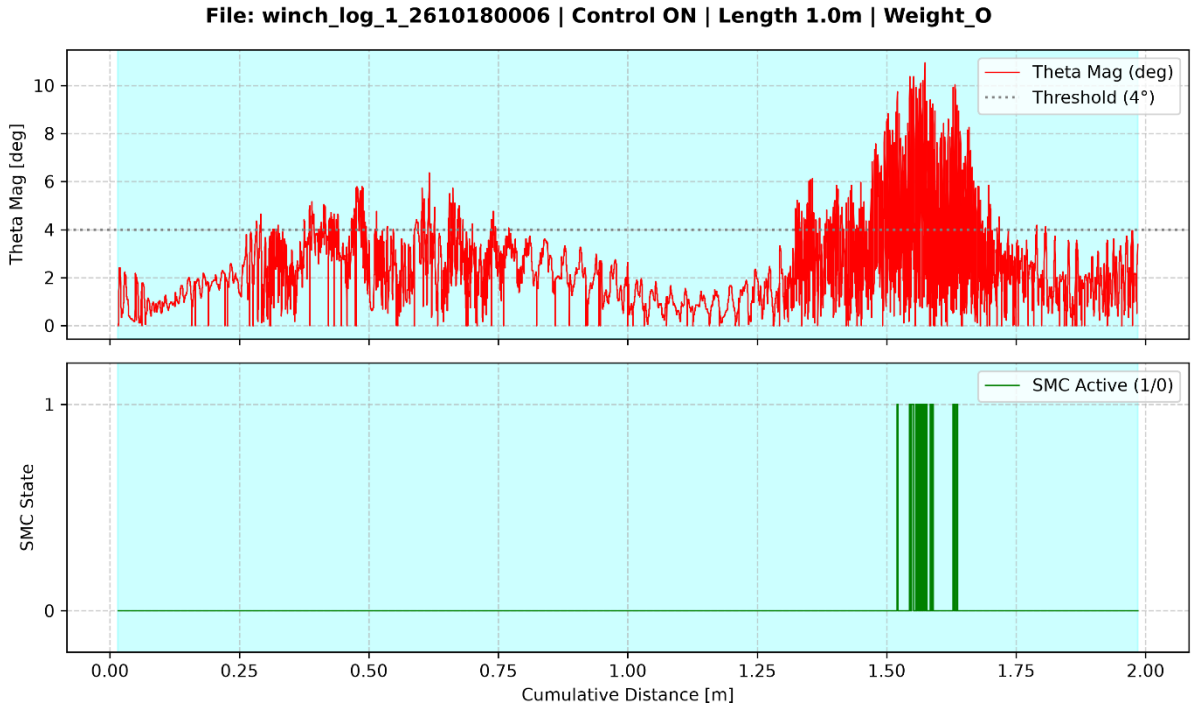

File: winch_log_1_2610180006 | Control ON | Length 1.0m | Weight_O
Theta Mag [deg]
Theta Mag (deg)
Threshold (4°)
SMC State
SMC Active (1/0)
Cumulative Distance [m]

Fig 37